

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

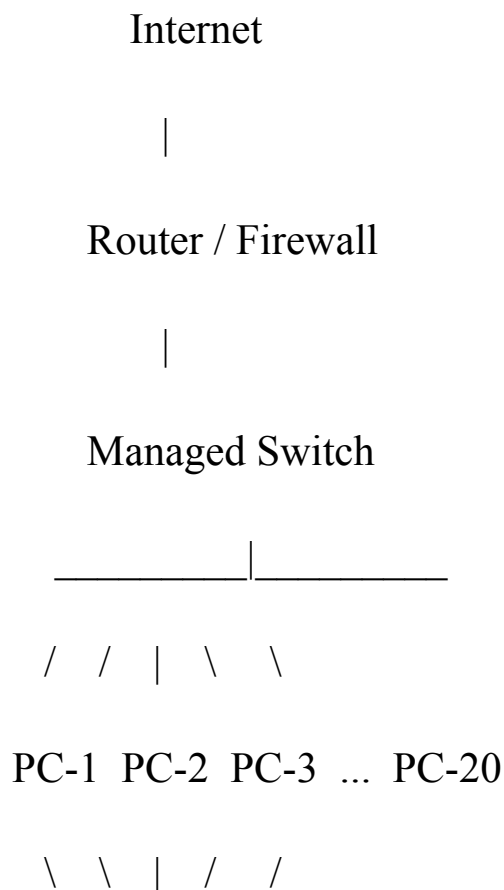
Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and

network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

sol: This describes a **star topology network design for a 20-person startup**. In simple terms:

Network Structure



Printer / VoIP / Wi-Fi AP

Key Points

1. **Central Switch:** A 24-port managed switch connects the 20 employees' devices and leaves some ports for future expansion.
2. **Cat6 UTP Cables:** Each device is connected to the switch using Ethernet cables, with the standard Ethernet copper channel generally limited to **100 m**.
3. **Router/Firewall:** Provides Internet connectivity and can handle services such as routing, DHCP, and NAT.
4. **Wi-Fi Access Point:** Connects wireless laptops, smartphones, and other mobile devices to the network.
5. **Fault Isolation:** If one cable or workstation fails, the remaining devices continue working because each device has its own connection to the switch.
6. **Easy Management:** Since all connections meet at the central switch, troubleshooting, maintenance, and adding new devices are easier.

Conclusion

A **star topology** is a suitable choice for a **20-person startup** because it provides centralized management, easy troubleshooting, good scalability, and fault isolation. The main disadvantage is that failure of

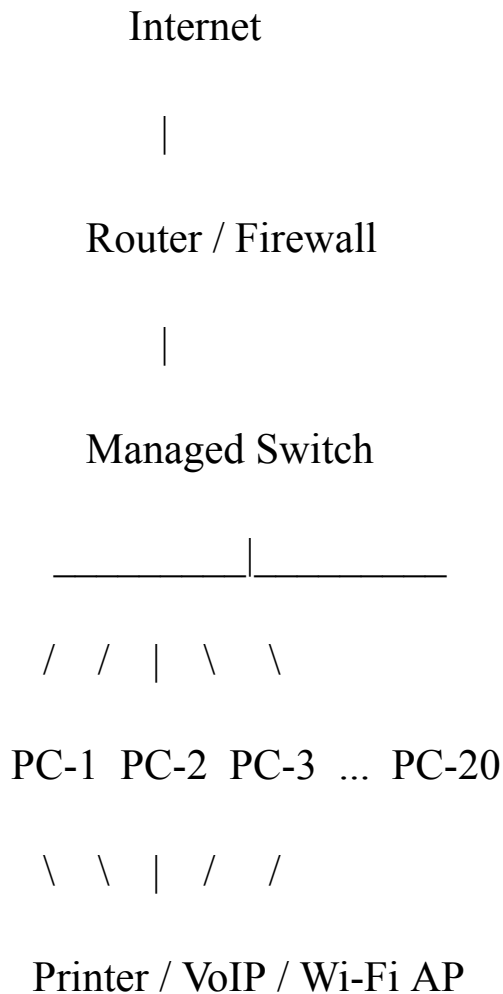
the **central switch** can affect all connected devices, so using a reliable managed switch and appropriate backup planning is important.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds

using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

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3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

This describes a **low-cost bus topology for a small classroom lab**. Here is a simple explanation:

Bus Topology Design

50Ω Terminator

 $-\cdot +$

Common Backbone Cable

Key Points

1. **Low Cost:** All computers share one common backbone cable, so less networking hardware is required.
2. **Coaxial Cable:** Traditional bus networks use **50-ohm coaxial cable**, with **terminators at both ends** to reduce signal reflections.
3. **T-Connectors:** Each computer is connected to the backbone using a T-connector in a classic coaxial Ethernet setup.
4. **Shared Medium:** All computers use the same communication channel, so simultaneous transmissions can cause collisions.
5. **Easy for Small Labs:** It can be suitable for a small classroom with light network traffic and a tight budget.
6. **Major Disadvantage:** A break in the backbone can disrupt communication for the entire network.

Important Correction

A modern Ethernet network using an 8- or 16-port unmanaged switch and Cat5e/Cat6 UTP is technically a star topology, not a bus topology. Each computer has a separate physical link to the switch.

So:

7. Traditional Bus:

8. PC1 --- PC2 --- PC3 --- PC4

9. Common Cable

10.

11. Modern Switched Ethernet:

12. Switch

13. / | | \

PC1 PC2 PC3 PC4

The second design may provide a **similar low-cost classroom solution**, but its physical topology is **star**, not bus.

Conclusion

For a **traditional bus topology**, coaxial cable, T-connectors, and terminators are required. It has low hardware cost but poor fault tolerance and shared-medium limitations. For a new classroom

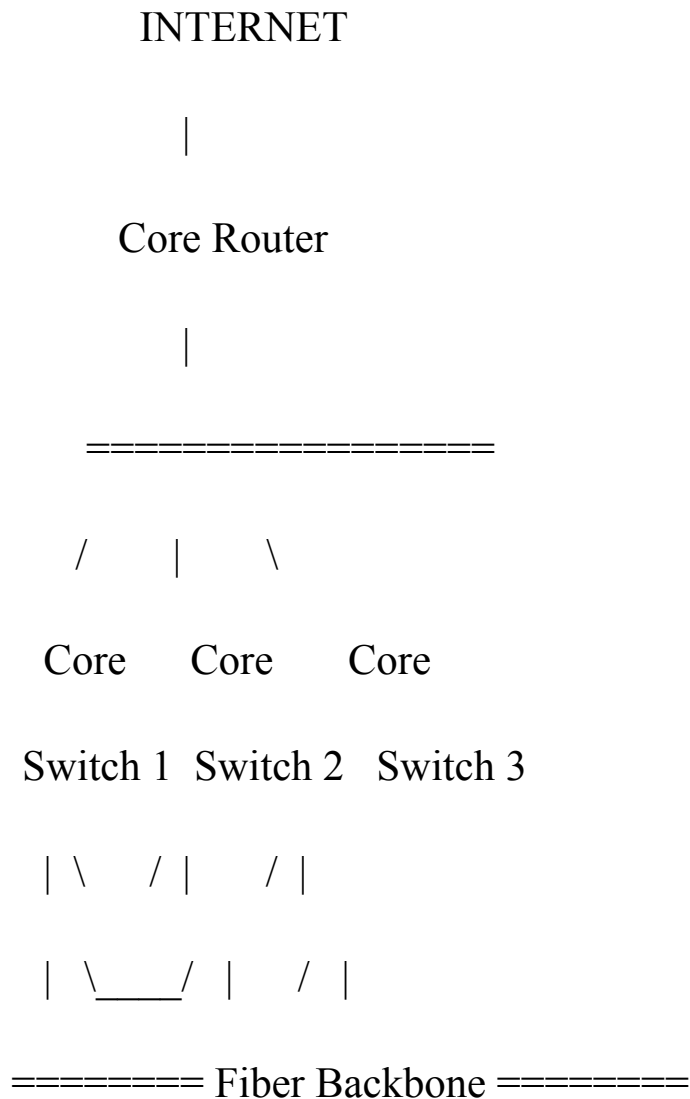
installation, **switched star topology is generally more practical and reliable**, even if the initial hardware cost is slightly higher

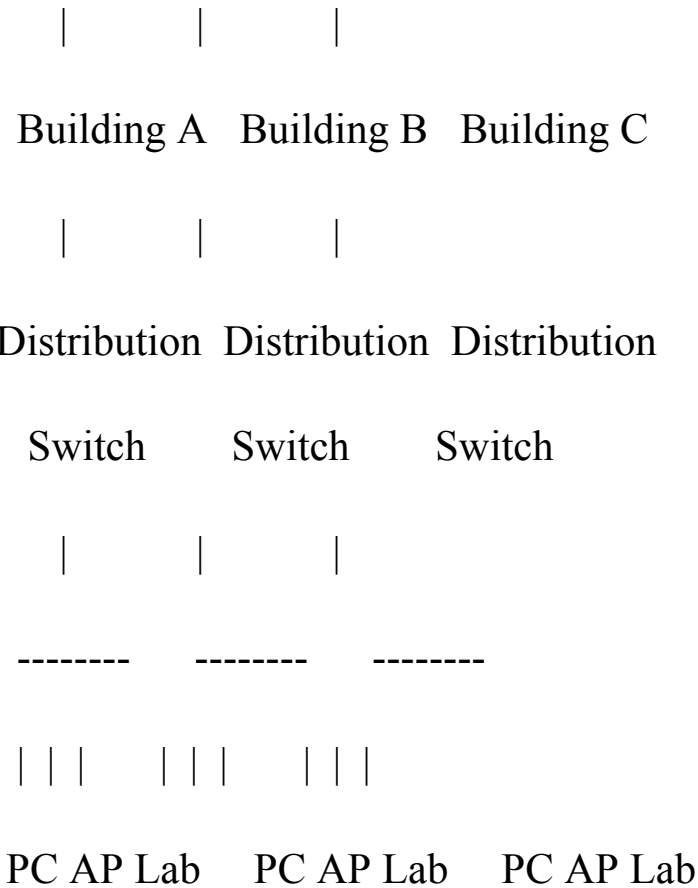
4 A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus

delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

This describes a **hybrid topology for a large university campus**. It combines different network topologies at different levels to achieve **reliability, manageability, and scalability**.

Hybrid Campus Network





Key Points

1. **Campus Backbone – Ring/Mesh:** High-speed fiber connects core routers and multilayer switches with redundant paths. If one link fails, traffic can use another path.
2. **Building Level – Star:** Each building has a distribution switch, with access switches connecting computers, printers, phones, and wireless access points.
3. **Lab Level:** Smaller or simpler structures can be used where appropriate for cost efficiency.

4. **Fiber Optic Links:** Single-mode fiber provides high-speed connections between buildings and the campus core.
5. **Dual Uplinks:** Buildings can have two fiber connections to improve redundancy and reduce the impact of a single link failure.
6. **PoE Wireless Access Points:** Access switches can provide both network connectivity and power to compatible wireless access points through Ethernet.
7. **Three-Layer Architecture:** The **core, distribution, and access** layers provide organized network management and scalability.

Why Hybrid Topology Is Suitable

Requirement	Hybrid Topology Solution	
Large campus	Multiple	interconnected networks
Reliability	Ring/mesh backbone	
Easy management	Star topology inside buildings	
High speed	Fiber-optic backbone	

Scalability Add buildings through
distribution switches

Wireless PoE access points
connectivity

Fault tolerance Redundant fiber links

Conclusion

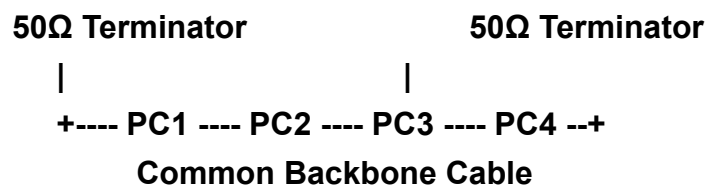
A **hybrid topology** is well suited to a university campus because different areas have different networking requirements. The **fiber ring/mesh provides backbone redundancy**, the **star topology simplifies building-level management**, and smaller local structures can reduce costs where appropriate. This combination provides a network that is **reliable, scalable, manageable, and suitable for thousands of users**.

5.Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices

like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

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The second design may provide a similar low-cost classroom solution, but its physical topology is star, not bus.

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For a traditional bus topology, coaxial cable, T-connectors, and terminators are required. It has low hardware cost but poor fault tolerance and shared-medium limitations. For a new classroom installation, switched star topology is generally more practical and reliable, even if the initial hardware cost is slightly higher.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution